

■■ SAFAR — Complete System Reference

Safety Assisting Forward-looking AI Reflex: Architecture, Pipeline & Variable Specification

1. Executive Overview & Design Philosophy

SAFAR is an advanced, physics-aware, AI-assisted road safety and collision avoidance subsystem designed for realistic vehicle simulation environments (Unreal Engine 5 Chaos Physics) and next-generation Advanced Driver Assistance Systems (ADAS).

Core Operating Principles:

- **Passive Driver Principle:** The driver maintains 100% authoritative control by default. SAFAR silently monitors trajectories and stopping limits, intervening only upon confirmed physical hazards.
- **Decoupled Perception & Reasoning:** Machine Learning models answer strictly *'What is this surface/object?'*, while deterministic kinematics, corridor geometry, and stopping equations determine physical intervention necessity.
- **Failure-Safe Guarantee:** Unconfident, missing, or invalid sensor inputs automatically evaluate to PASSIVE without false emergency braking.

2. End-to-End Safety Pipeline Flow

Pipeline Stage	Component	Input → Output	Function
1. Sensing	Virtual Stereo & YOLO	Raw Camera / World Actors → Bounding Boxes & Disparity	Detects vehicles, pedestrians, animals, and waterlogged road depressions.
2. Validation	PotholeDataValidator	Raw Dimensions → Validated Measurements	Rejects NaN, Inf, and negative values to prevent crashes.
3. Tracking	Kinematic Tracker	Positions over Time → Relative Velocity (V_{rel})	Computes 60Hz velocity vectors and dead reckoning.
4. Trajectory	Corridor Predictor	V_{rel} & Speed → Corridor Intersection & TTC	Filters objects outside vehicle corridor ($ Y \leq 1.05m$).
5. Physics	Kinematics Engine	Speed & Deceleration → d_{stop} & Safety Ratio	Computes stopping distance $d = v \cdot t + v^2 / (2a)$.
6. Decision	State Machine	Risk Score → Vehicle Command	Hysteresis (0.70/0.40), ≥ 2 -frame confirmation & 0.35s hold.
7. Actuation	Chaos Vehicle Actuator	Command → Brake / Throttle	Speed-gated service braking vs stationary handbrake lock.

3. File-by-File Technical Summary

- **safar/pothole/config.py:** Central configuration defining physical constants ($t_{react}=0.18s$, $a_{nom}=6.0 \text{ m/s}^2$, $a_{emg}=8.5 \text{ m/s}^2$), lane half-widths (1.75m), vehicle envelope (1.05m), safe speed maps, and activation/release thresholds.
- **safar/pothole/validation.py:** Data sanitization layer with PotholeDataValidator verifying that dimensions are strictly positive, non-null, finite numbers within valid bounds ($W \leq 10m$, $L \leq 20m$, $D \leq 1.0m$).
- **safar/pothole/model.py:** ML trainer and serialization engine using Stratified 5-Fold Cross-Validation, evaluating Decision Tree, Random Forest, Extra Trees, and Gradient Boosting before exporting pothole_model.joblib.
- **safar/pothole/classifier.py:** Decoupled classification answering only *'What surface is this?'*. Employs Gradient Boosting with calibrated softmax probabilities, outputting PotholeObservation and flagging confidence < 0.70 as UNCERTAIN.
- **safar/pothole/physics.py:** Kinematics engine computing dynamic stopping envelope $d_{stop} = v \cdot t_{react} + v^2 / (2a)$, required stopping buffer, safety ratio $R = d / d_{stop}$, and safe time-to-pothole $t = d / v$.

- **safar/pothole/path.py**: Corridor geometry tracking lateral offset X relative to vehicle centerline, classifying geometry into PATH_CLEAR, POSSIBLE_INTERSECTION, and INTERSECTION.
- **safar/pothole/risk.py**: Transparent physical risk engine synthesizing depth severity, approach speed excess, stopping distance ratio, and lateral relevance into continuous danger score [0.0, 1.0].
- **safar/pothole/decision.py**: Finite state machine (MAINTAIN -> MONITOR -> SLOW -> BRAKE -> EMERGENCY_BRAKE) with temporal confirmation (≥ 2 frames) and 0.35s anti-flapping hold timers.
- **safar/pothole/simulation.py**: Multi-hazard aggregator evaluating all detected anomalies in a single frame and selecting primary threat deterministically.
- **safar/perception/yolo_detector.py**: Ultralytics YOLOv8 wrapper performing real-time multi-class road obstacle detection (cars, trucks, motorcycles, pedestrians, dogs) with ego hood rejection.
- **tools/test_safar_on_images.py**: End-to-end evaluation pipeline running YOLO + Optical Pothole Segmentation + Kinematics on real-world images with visual HUD overlays and JSON logs.
- **Source/TrafficGame/SAFAR/**: Unreal Engine 5 C++ closed-loop driver safety core with Chaos Wheeled Vehicle Movement integration, stereo depth rig, and automatic transmission reverse protection.

4. Exhaustive Variable & Parameter Dictionary

Variable Name	Type / Units	Default	Defined In	Physical / Functional Meaning
REACTION_TIME_S	float (s)	0.18	config.py	Total latency: sensor acquisition + perception + brake pressurization.
DECEL_NOMINAL_MPS2	float (m/s ²)	6.0	config.py	Nominal service braking deceleration comfort limit on dry pavement.
DECEL_EMERGENCY_MPS2	float (m/s ²)	8.5	config.py	Maximum deceleration under full anti-lock emergency braking.
CONFIDENCE_THRESHOLD	float [0, 1]	0.70	config.py	Minimum ML probability required; lower values flagged as UNCERTAIN.
ACTIVATION_THRESHOLD	float [0, 1]	0.70	config.py / Decision	Threat risk score required to engage active automated braking.
RELEASE_THRESHOLD	float [0, 1]	0.40	config.py / Decision	Lower risk score boundary required before releasing brakes (hysteresis).
MIN_HOLD_DURATION_S	float (s)	0.35	config.py / Decision	Anti-flapping hold timer maintaining brake pressure after obstacle clears.
THREAT_CONFIRMATION_FRAMES	int (frames)	2	config.py / Decision	Consecutive frames obstacle must pose a threat before braking activates.
VEHICLE_HALF_WIDTH_M	float (m)	1.05	config.py / path.py	Physical half-width of vehicle track plus safety buffer.
LANE_HALF_WIDTH_M	float (m)	1.75	config.py / path.py	Standard driving lane half-width.
CAMERA_FOCAL_LENGTH_PX	float (px)	720.0	test_safar_on_images	Focal length used in monocular pinhole distance estimation.
CAMERA_MOUNT_HEIGHT_M	float (m)	1.35	test_safar_on_images	Height of windshield-mounted ADAS camera above ground plane.
STEREO_BASELINE_M	float (m)	0.25	SAFARSensorRig	Physical distance between virtual stereo camera lenses in UE5.
REVERSE_PREVENTION_SPEED	float (m/s)	0.5	SAFARVehicleComp	Speed cutoff where service brake is replaced with handbrake lock.

5. Key Mathematical Formulations

1. Dynamic Stopping Distance: $d_{\text{stop}} = v * t_{\text{react}} + v^2 / (2 * a)$

2. Time-To-Collision (TTC): $TTC = d_{\text{longitudinal}} / V_{\text{closing}}$ where $V_{\text{closing}} = v_{\text{ego}} - v_{\text{target}}$

3. Metric Stereo Depth: $Z = (f * \text{Baseline}) / \text{disparity}$

4. Pinhole Monocular Distance: $Z = (f * \text{Real_Height}) / \text{bbox_height_px}$ and $X = (\text{center_x} - c_x) * Z / f$

5. Pothole Risk Score: $R_{\text{risk}} = \text{BaseSeverity} * (0.65 * D_{\text{urgency}} + 0.35 * S_{\text{excess}}) * M_{\text{lateral}} * \text{Confidence}$